

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 2600



Problem Description:

For the second sequence of design activities, the framing of the design problem and solution approach will change. However, the same problem statement and performance model are used. Continue playing the role you were assigned in the last round.

The original problem statement is reprinted here for your reference:

You are playing the role of a design engineer working for a company that develops military ground vehicles. The US Army seeks a replacement for the HMMWV ("Humvee"), which has been in service since 1985. Your company is tasked with developing a replacement vehicle. The HMMWV fills a light-duty, all-purpose role, and similarly, its replacement requires a balance of performance along multiple dimensions.

The company you work for is divided into two working groups: the **Suspension and Driveline** working group, and the **Chassis and Cabin** working group. The Suspension and Driveline working group is concerned with maximizing the on- and off-road mobility of the vehicle, while the Chassis and Cabin working group is concerned with the packaging, protection, and interior usability of the vehicle. Your company uses a model that predicts how a ground vehicle performs in 10 dimensions of performance. The model takes 10 design variables as input. The design variables and performance dimensions are divided amongst the working groups, detailed below.

Definition of Terms:

Performance dimensions serve as design objectives that your company wishes to maximize. Each working group is especially concerned with maximizing the objectives that fall within their discipline.

Suspension and Driveline Performance Dimensions:

A Top Speed: Maximum vehicle speed achievable on flat, paved surfaces

B Acceleration: Rate of increase of vehicle speed from a standstill

Off-Road Ability: Ability of vehicle to traverse non-flat, non-paved, low-traction environments

Operational Range: Maximum round-trip mission distance in a single fuel tank/battery charge

Towing Capacity: Ability of vehicle to tow equipment and other vehicles

Chassis and Cabin Performance Dimensions:

Low-Speed Maneuverability: Ability of vehicle to drive in confined environments

Occupant Protection: Ability of vehicle to protect occupants from road hazards and combatants

C Maintainability: Ease of performing maintenance and robustness to failures requiring repair

D Rollover Stability: Ability of vehicle to handle abrupt, high-speed steering inputs without rollover

Passenger Cabin Space: Usable interior volume of vehicle

PROBLEM DESCRIPTION CONTINUES ON NEXT PAGE

Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

107in	98.12in	66.6in	23.1in	512lb-ft	5.32:1	30in	1530kWh	0.284in	0.84in
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Do you think you could have found a better solution given more time?

Yes

What, in this round, stopped you from finding a better solution?

Time, lack of Information + understand
wants necessary for military
vehicles specifically

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

107in	87.6in	78in	8.1in	750lb-ft	2.93:1	36in	880kWh	0.335in	1.97in
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Do you think you could have found a better solution given more time?

Yes.

What, in this round, stopped you from finding a better solution?

Time, and lack of requirements

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

107in	98.2in	66.6in	23.1in	5126-4	5.32:1	30in	1530Kwh	0.284in	0.84in
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Do you think you could have found a better solution given more time?

Yes

What, in this round, stopped you from finding a better solution?

Nothing, we believe we have found the best choice

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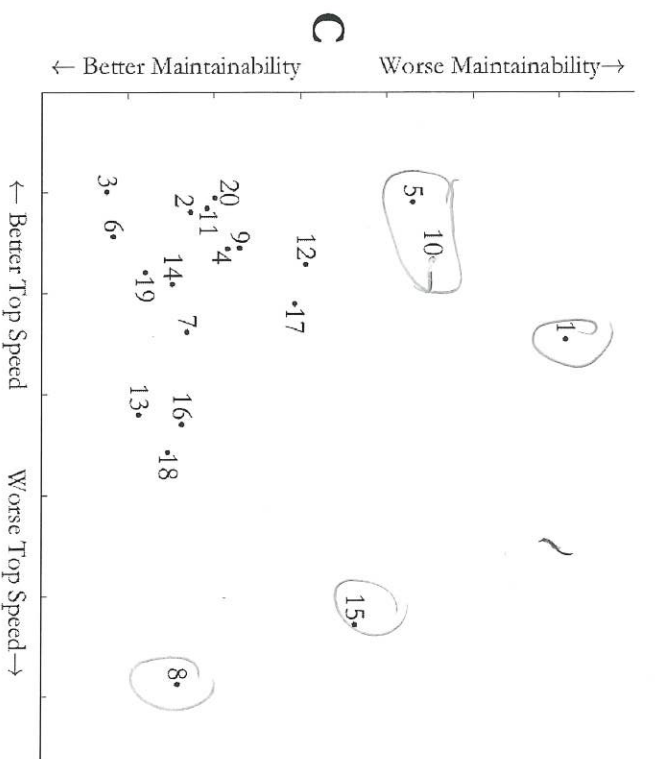
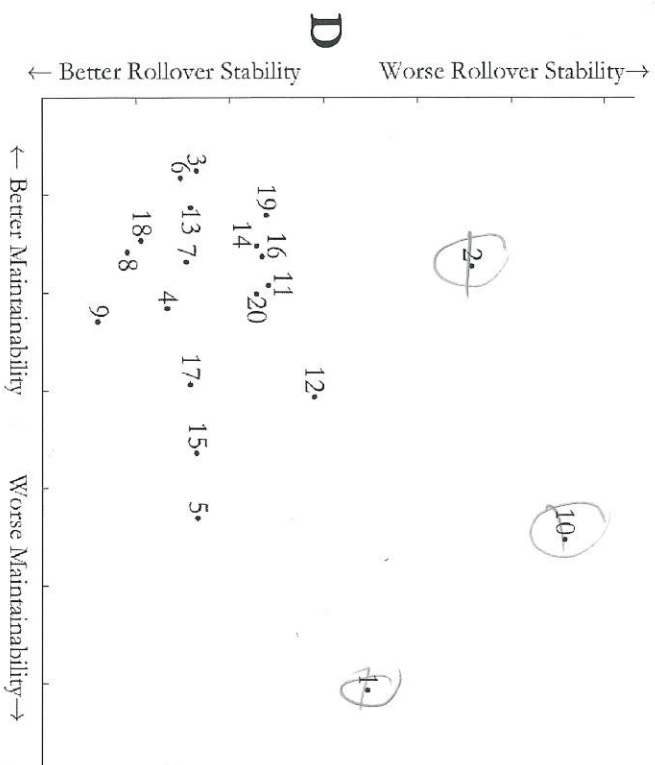
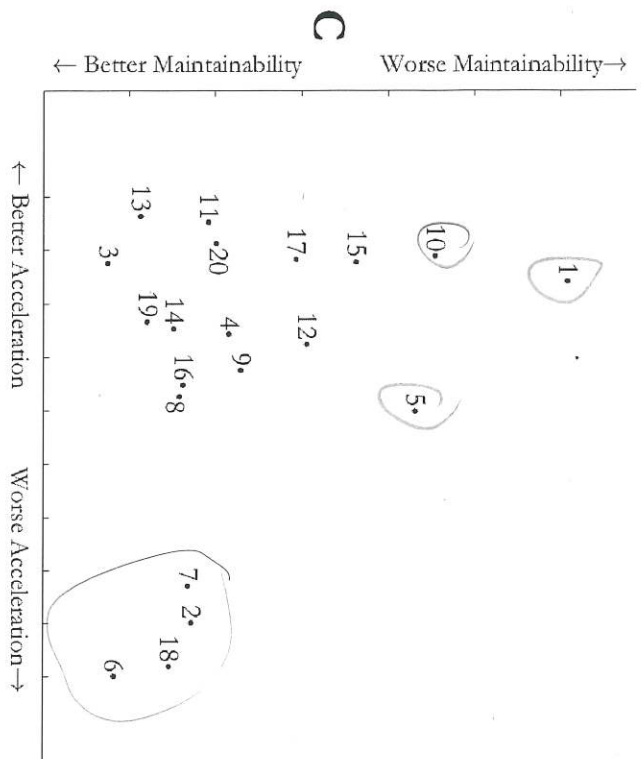
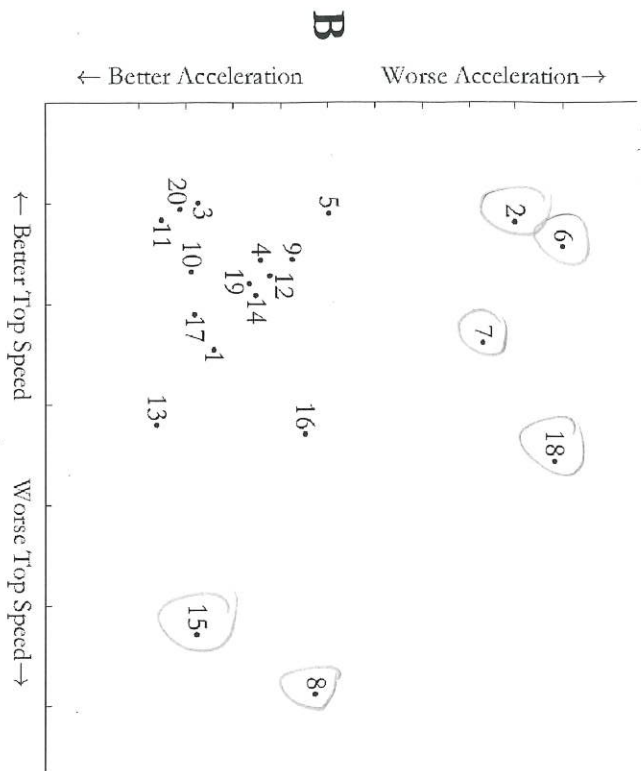
Vehicle Index

	Trackwidth	Wheelbase	Roof Height	Suspension Travel	Max Enging Torque	Driveline Gearing	Tire Diameter	Energy Capacity	Frame Rail Wall Thickness	Body Panel Thickness
1	196 in	80.2 in	79.3 in	22.9 in	728 lb-ft	5.87:1	46 in	1620 kWh	0.948 in	0.98 in
	Very Wide	Very Short	Tall Roof	Very Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Moderately Armored
2	110 in	80.1 in	65.5 in	20.8 in	421 lb-ft	2.15:1	49 in	1660 kWh	0.939 in	1.48 in
	Very Narrow	Very Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Heavily Armored
3	119 in	80.2 in	70.2 in	6.8 in	543 lb-ft	4.67:1	48 in	1560 kWh	0.285 in	0.18 in
	Very Narrow	Very Short	Short Roof	Very Short Travel	Moderate Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
4	181 in	81.2 in	69.2 in	9.7 in	655 lb-ft	2.37:1	32 in	1070 kWh	0.877 in	1.99 in
	Very Wide	Very Short	Short Roof	Short Travel	High Output	Very Long Gearing	Very Small Tire	Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
5	192 in	85.6 in	65.3 in	21.1 in	684 lb-ft	2.61:1	45 in	1240 kWh	0.783 in	1.48 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very High Output	Very Long Gearing	Large Tire	Moderate Energy Capacity	Heavy-Duty Frame	Heavily Armored
6	178 in	81.1 in	79.1 in	6.8 in	409 lb-ft	2.53:1	47 in	1530 kWh	0.251 in	0.47 in
	Wide	Very Short	Tall Roof	Very Short Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
7	199 in	84.2 in	68.2 in	22.7 in	419 lb-ft	2.4:1	39 in	1520 kWh	0.331 in	1.74 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Moderate Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
8	176 in	98.8 in	69.2 in	10.8 in	443 lb-ft	5.08:1	31 in	960 kWh	0.994 in	1.67 in
	Wide	Very Long	Short Roof	Short Travel	Very Low Output	Short Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
9	195 in	96.4 in	66.1 in	9.3 in	630 lb-ft	5.12:1	50 in	1610 kWh	0.741 in	0.17 in
	Very Wide	Very Long	Very Short Roof	Very Short Travel	High Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
10	134 in	83.1 in	80 in	18.1 in	706 lb-ft	5.95:1	49 in	1010 kWh	0.512 in	1.75 in
	Narrow	Very Short	Tall Roof	Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very Low Energy Capacity	Light-Duty Frame	Very Heavily Armored
11	107 in	87.6 in	78 in	8.1 in	750 lb-ft	2.93:1	30 in	880 kWh	0.335 in	1.97 in
	Very Narrow	Short	Tall Roof	Very Short Travel	Very High Output	Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
12	114 in	99.7 in	82 in	18.4 in	715 lb-ft	2.25:1	30 in	980 kWh	0.967 in	0.74 in
	Very Narrow	Very Long	Very Tall Roof	Long Travel	Very High Output	Very Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
13	107 in	98.2 in	66.6 in	23.1 in	512 lb-ft	5.32:1	30 in	1530 kWh	0.284 in	0.84 in
	Very Narrow	Very Long	Very Short Roof	Very Long Travel	Low Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
14	110 in	92 in	67.4 in	20.6 in	517 lb-ft	2.84:1	33 in	1210 kWh	0.432 in	1.82 in
	Very Narrow	Long	Very Short Roof	Very Long Travel	Low Output	Long Gearing	Very Small Tire	Moderate Energy Capacity	Light-Duty Frame	Very Heavily Armored

Vehicle Index

15	200 in	95.1 in	81.7 in	20.6 in	665 lb-ft	5.92:1	31 in	1460 kWh	0.313 in	1.67 in
	Very Wide	Long	Very Tall Roof	Very Long Travel	High Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
16	124 in	99.8 in	82.1 in	18.8 in	532 lb-ft	3.62:1	31 in	1570 kWh	0.782 in	0.33 in
	Narrow	Very Long	Very Tall Roof	Long Travel	Low Output	Moderately Gearing	Very Small Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
17	172 in	94 in	83.4 in	9.6 in	735 lb-ft	5.94:1	43 in	1500 kWh	0.313 in	0.78 in
	Wide	Long	Very Tall Roof	Very Short Travel	Very High Output	Very Short Gearing	Large Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
18	192.00527	98 in	82.4 in	6.5 in	442 lb-ft	3.23:1	35 in	1120 kWh	0.308 in	1.7 in
	Very Wide	Very Long	Very Tall Roof	Very Short Travel	Very Low Output	Long Gearing	Small Tire	Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
19	108 in	84.8 in	66.3 in	9.6 in	456 lb-ft	5.74:1	50 in	1080 kWh	0.966 in	0.84 in
	Very Narrow	Short	Very Short Roof	Very Short Travel	Very Low Output	Very Short Gearing	Very Large Tire	Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
20	102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
	Very Narrow	Long	Moderate Height Roof	Very Short Travel	Very High Output	Short Gearing	Large Tire	Very Low Energy Capacity	Heavy-Duty Frame	Very Lightly Armored

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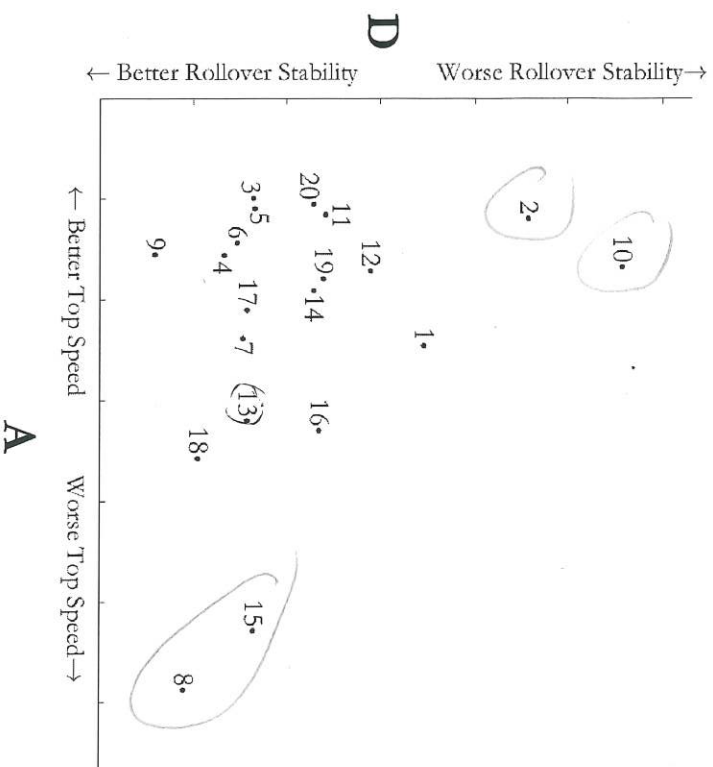
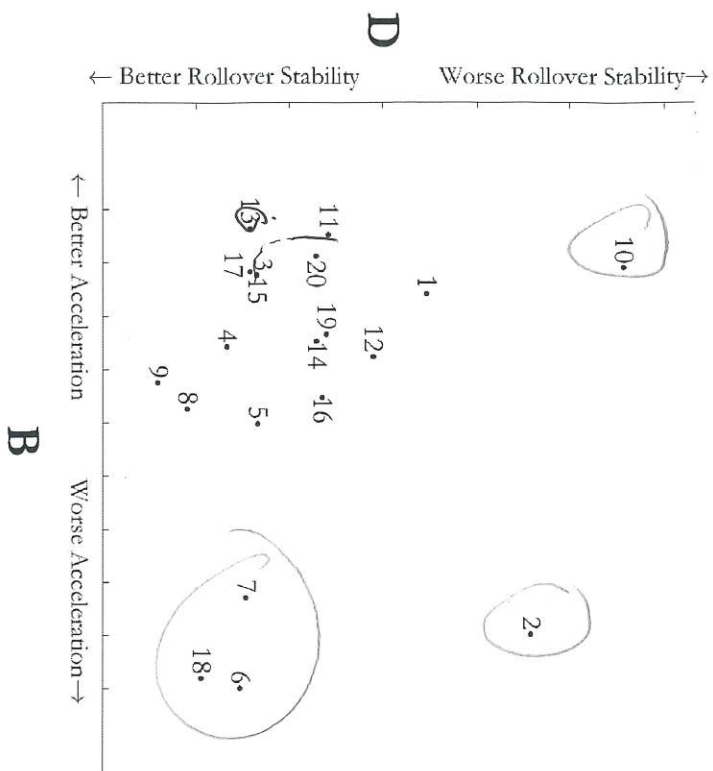
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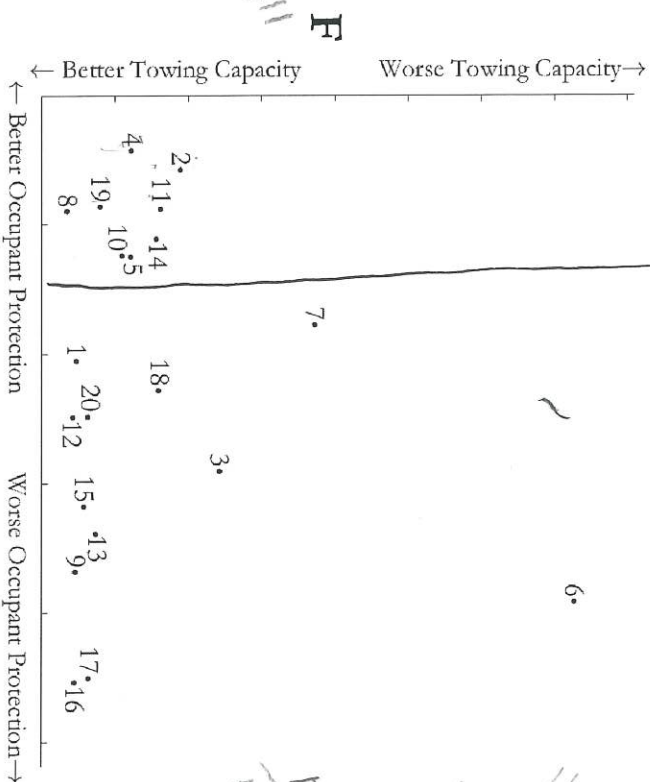
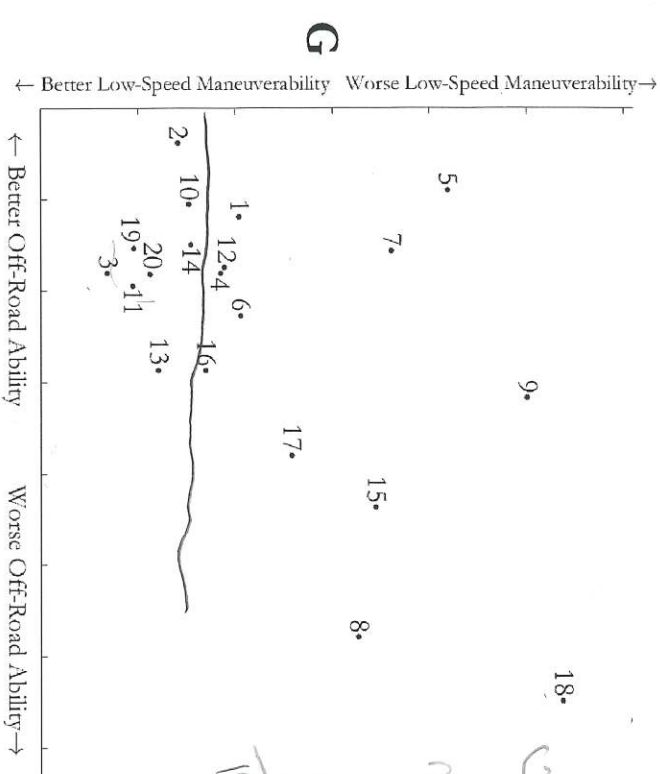
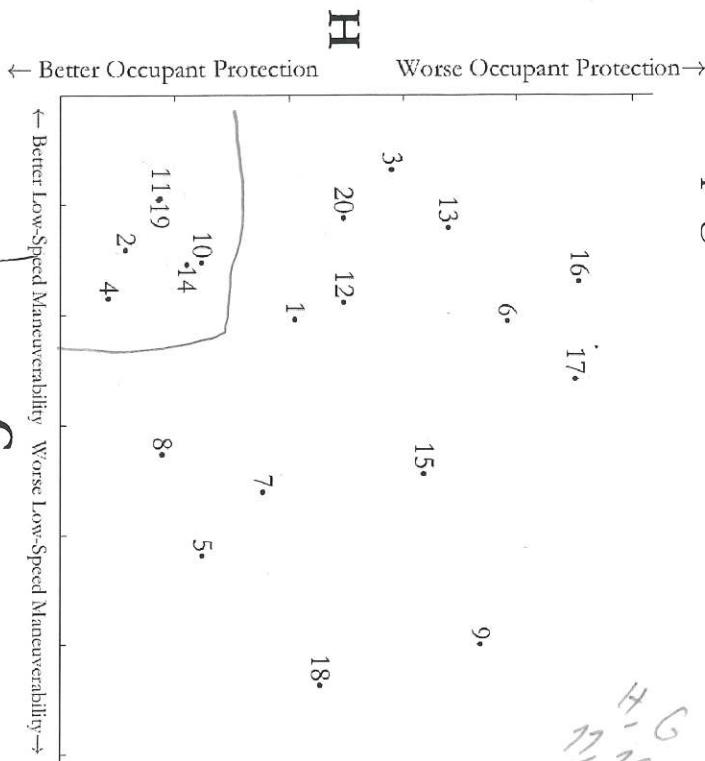
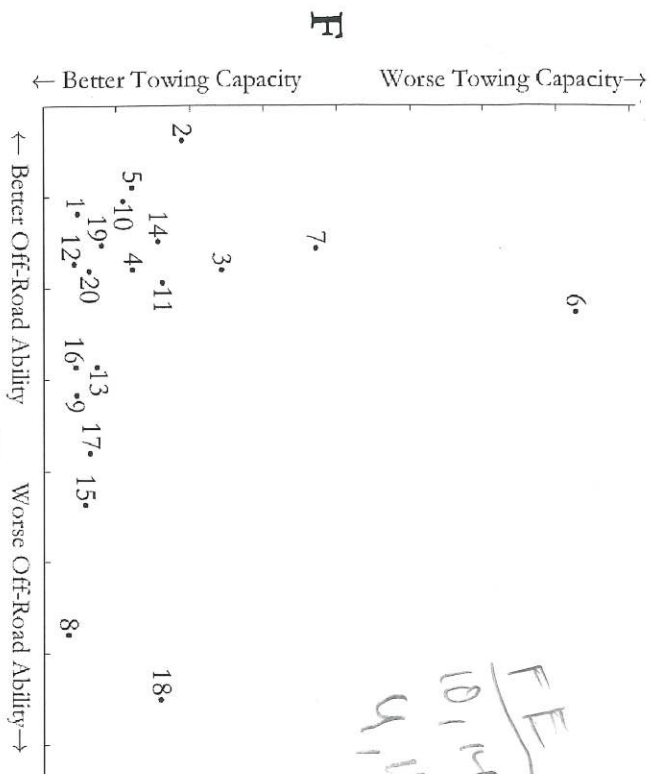
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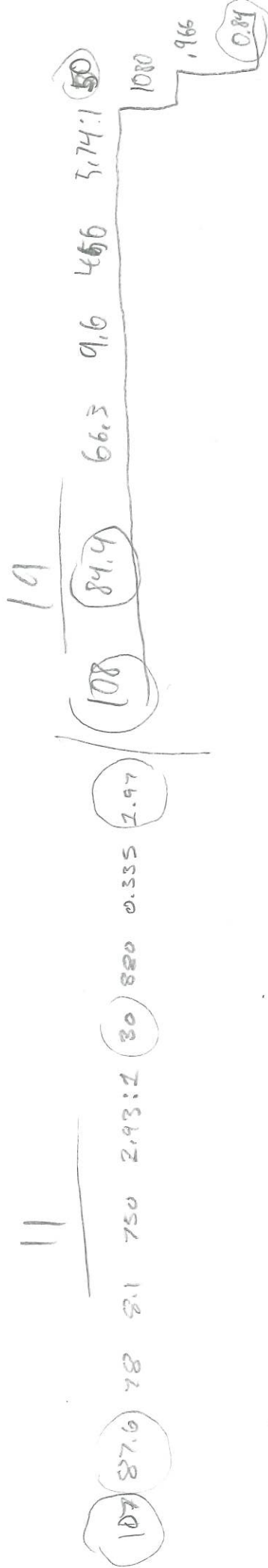


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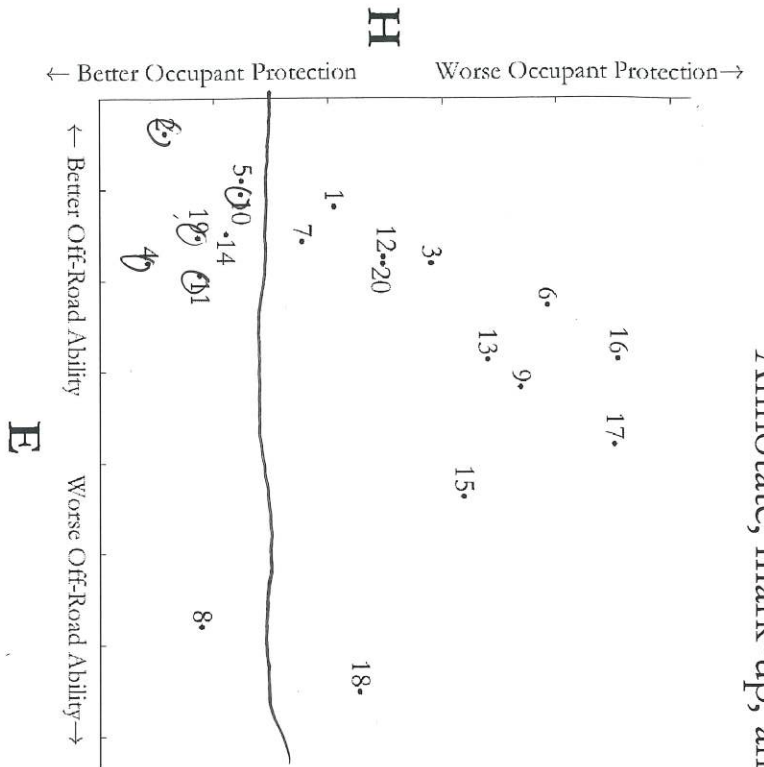
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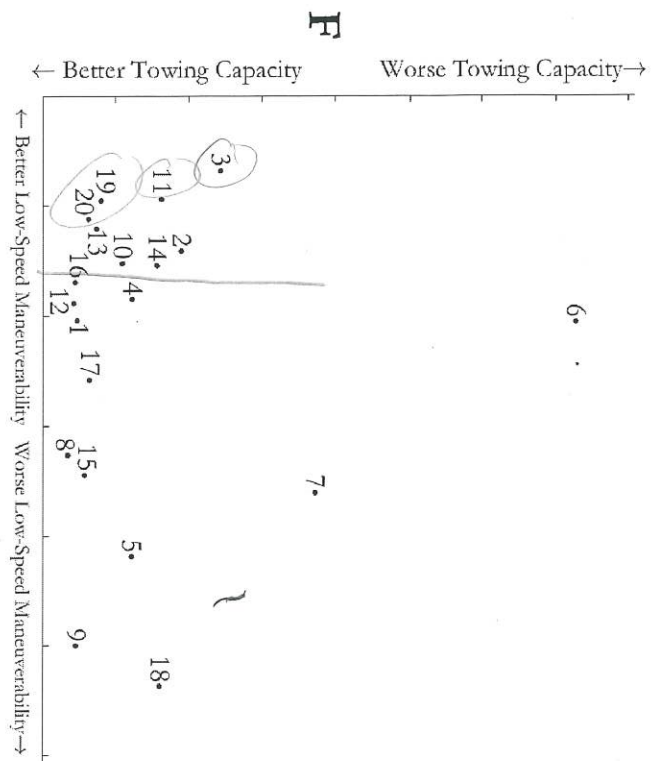




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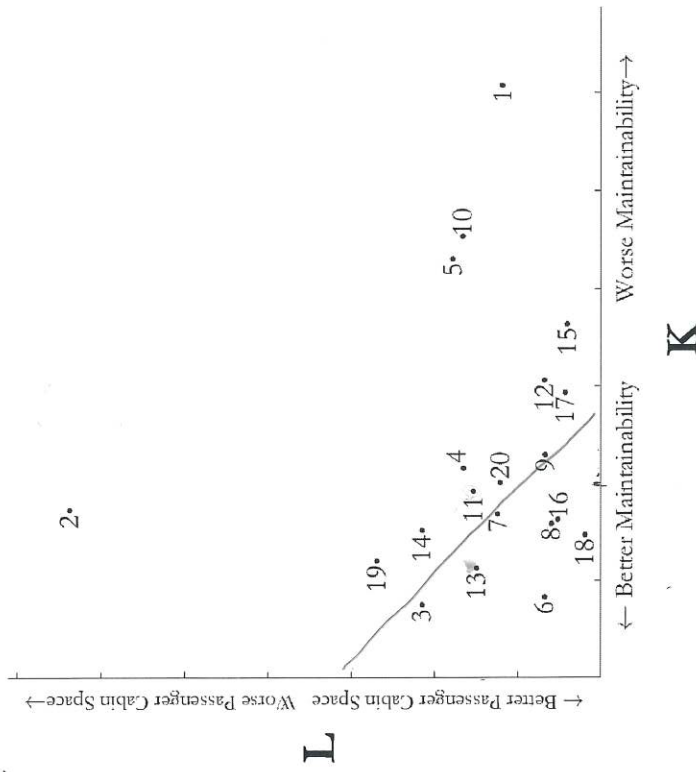
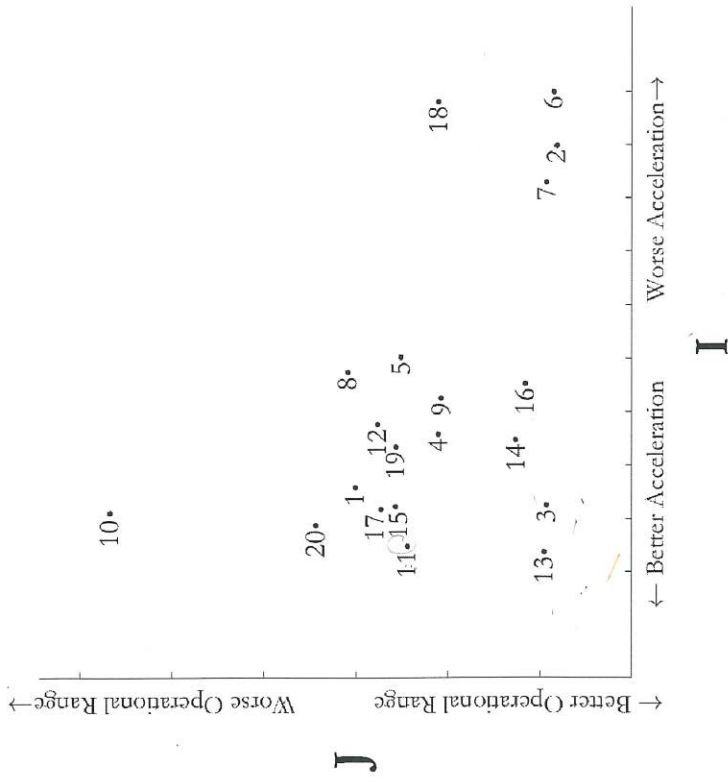
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